

TRINARY SYMMETRY

From the 360-Degree Arithmetic Wheel to Eisenstein-Lattice Type-II Analysis

A Comprehensive Working Manuscript

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Working research manuscript - not a claim of a proof of the Riemann Hypothesis

Abstract

This manuscript consolidates the Trinary Symmetry research program from its earliest 360-degree modular wheel and dual $6n-1 / 6n+1$ prime-track picture through the later cone, lattice, Eisenstein-integer, and Type-II angular-sieve developments. Arithmetic invariants are separated from geometric embeddings, computational observations, conjectures, and conditional analytic programs. The appendices collect the proof chain developed to date.

The original August 2026 operator/RH manuscript is retained as a conditional historical stage. The later analytic program is organized around bilinear angular sums in the Eisenstein integers, determinant strips, content decomposition, Smith normal form, prime-ideal extraction, additive large-sieve input, recursive scale descent, and a weighted multiplicative-energy estimate for the exact diagonal.

Status convention

Label	Meaning
PROVED	Proof or derivation included in the appendices.
COMPUTED	Finite computational observation only.
MODEL	Geometric visualization or embedding.
CONJECTURE	Proposed but unproved relation.
CONDITIONAL	Valid under stated hypotheses.
AUDIT	Proof architecture requiring independent verification.

Part I. The 360-Degree Arithmetic Wheel

1. Residue coordinates and 360 spokes

MODEL: The wheel is an exact residue bookkeeping device with a chosen circular embedding.

Every integer x has a unique decomposition $x=360q+r$ with $0\leq r<360$. Placing r at angular coordinate r degrees creates a 360-spoke wheel while q records the quotient or shell. Because 360 has many small divisors, the wheel simultaneously displays numerous low-modulus congruence structures. The arithmetic object is the residue coordinate; the Euclidean circle is a representation.

$$x = 360 q + r, \quad r \text{ in } \mathbb{Z}/360\mathbb{Z}.$$

2. Fixed prime tracks modulo 6

PROVED: Every prime $p>3$ lies on exactly one of the tracks $6n-1$ or $6n+1$.

A prime is a fixed integer and therefore has a fixed residue class. Earlier track-switching language was discarded. Density fluctuations and composite obstruction patterns may change with scale, but an individual prime never switches tracks.

$$\tau_{-}=\{6n-1\}; \quad \tau_{+}=\{6n+1\}.$$

3. Signed-track multiplication

PROVED: The tracks carry an exact $\mathbb{Z}_2$ multiplication law.

$$(6a+s)(6b+t)=6(6ab+at+bs)+st, \quad s, t \text{ in } \{-1, +1\}.$$

The output track sign is st and the quotient coordinate is $K=6ab+at+bs$. This exact algebra is the arithmetic core behind the antipodal picture.

4. Antiparallel phase and contraction

PROVED: The real character χ_6 is multiplicative on integers coprime to 6.

$$\chi_6=+1 \text{ on } 1 \bmod 6; \quad -1 \text{ on } 5 \bmod 6. \quad W(x)=\chi_6(x)/\sqrt{x}.$$

For $p=5 \bmod 6$, $W(p^k)=(-1)^k p^{(-k/2)}$; for $p=1 \bmod 6$ the sign remains positive. The 180-degree visualization is faithful to this sign law, although any cone or sphere is an optional embedding.

5. Base-3 and base-2 roles

MODEL: Base choices are coordinate or sampling devices, not causes of primality.

Base 3 entered through the original trinary interval geometry; powers of 2 later served as empirical checkpoints for counts and density. The current formulation keeps the arithmetic base-invariant.

Part II. Geometric Embeddings and Topology

6. Cone and double-cone models

MODEL: The cone embeds quotient growth and residue angle.

Opposing or nested cones were used to visualize the two fixed tracks and mirror spokes. A conical helix can preserve reflection and antipodal symmetry, but it does not imply a literal finite intersection of the arithmetic tracks.

7. Plane, sphere, spiral, and zero state

MODEL: These are compactification and inversion pictures.

The project explored a cone expanding toward an infinite plane, followed by an inward spherical spiral and return toward zero. The rigorous replacements for these pictures are reciprocal scaling, residue characters, projective coordinates, and inverse-limit topology.

8. Nested residues

PROVED: Compatible residues modulo 6^k form a profinite inverse system.

The normalized Euclidean gap $2/6^k$ tends to zero, but the natural inverse-limit topology is ultrametric. Chinese remainder structure separates the 2-adic and 3-adic components. This distinguishes normalized closeness from literal arithmetic intersection.

9. Fixed positions before waves

METHODOLOGY: Prime positions are fixed; wave language is not used to establish the structure.

Multiplication and divisibility propagate composite structure. Terms such as orbiters, prime holes, shell smoothing, and interference are exploratory unless translated into explicit algebra or analysis.

Part III. Eisenstein and Hexagonal Structure

10. Eisenstein integers

PROVED: $R=\mathbb{Z}[\omega]$ is Euclidean, hence a PID and UFD.

$$N(a+b\omega)=a^2-ab+b^2.$$

The six units generate sixfold rotational symmetry. Ideals or normalized generators remove unit ambiguity and make gcd/content decompositions precise.

11. Quadratic cone form

PROVED: $Q(q,r)=3q^2+3qr+r^2$ is invariant under $T(q,r)=(q,-3q-r)$, and T is an involution.

$$T^2=I; \quad Q(T(q,r))=Q(q,r).$$

The shell label $p=Q+41$ was computationally useful, but Q is not a one-to-one label for all integers. A global bridge from cone coordinates to $6m+/-1$ remains unproved.

12. Computational observations

COMPUTED: Finite scans show reproducible symmetry but do not establish global coverage.

Reported tests included 539/539 represented primes below roughly 100,042 satisfying the selected representation identity, multiple symmetric representations of 2069 including the anchor (52,-78), and six-residue count vectors with recurrence-like behavior. These are finite observations.

13. Original conditional operator framework

CONDITIONAL: The August 2026 operator/RH architecture is historical motivation, not an unconditional theorem.

That manuscript proposed base-3 interval geometry, a modulo-6 antiparallel phase engine, Sobolev and trace-class operators, adelic kernels, Fredholm determinants, and Galerkin discretization. It also reported a 98.08 percent finite-scan reduction in a structural-background metric. The present synthesis does not use those claims as an unconditional proof of RH.

Part IV. Angular Type-II Program

14. Bilinear target

AUDIT: The proof must preserve the two Type-II variables.

$$T_M = \sum_{1 \leq |m| \leq M} \left| \sum_{N(u) \sim U, N(v) \sim V} a_u b_v \exp(6im \arg(uv)) W(N(u)N(v)/X) \right|^2.$$

$$\text{Target: } T_M \ll (M+X) X^\epsilon \left(\|a\|_{L^2}^2 \|b\|_{L^2}^2 \right), \quad UV=X.$$

A same-sequence product-set fourth moment was rejected after short-sector counterexamples. In the genuine bilinear form, concentration on one factor imposes a corresponding angular restriction on the other.

15. Determinant strips

PROVED: The Eisenstein determinant controls angular separation.

$$[z,w]=ad-bc; |\sin(\arg z-\arg w)|=\sqrt{3}|[z,w]|/(2\sqrt{Nz}\sqrt{Nw}).$$

After freezing three Type-II variables, the fourth variable lies in an Eisenstein annulus intersected with a strip $Px+Qy$ of controlled width.

16. Centered incidence lesson

PROVED: Subtracting the uniform angular mode is exact but insufficient for arbitrary concentrated coefficients.

$$A^o_{\{r,s\}}=A_{\{r,s\}}-\kappa_s M_r, \sum_s A^o_{\{r,s\}}=0.$$

This centering isolates the bulk angular-density term, but a short-sector example disproves the desired same-sequence centered fourth-moment estimate. This failure is one reason the bilinear structure is essential.

17. Primitive strip content

PROVED: $q=\gcd(P,Q)$ converts geometric boundary loss into arithmetic content.

$$P=qP_0, Q=qQ_0, \gcd(P_0, Q_0)=1, |P_0 x + Q_0 y| \leq H/q.$$

Primitive fibers are affine lattice lines. The crude square-root boundary term is reorganized into content-dependent terms, revealing a congruence problem rather than a purely geometric perimeter problem.

18. Small content

AUDIT: The required independent input is a two-dimensional additive large sieve.

$$\sum_{\mathbf{x}} |\sum_{\mathbf{z}} c_{\mathbf{z}} e(\mathbf{x} \cdot \mathbf{z})|^2 \ll (V+D^4) X^\epsilon \prod |c_{\mathbf{z}}|^2.$$

Primitive rational frequencies with denominators $d \leq D$ are separated at scale D^{-2} in the two-torus. Taking $D=V^{1/4}$ balances D^4 with V . The remaining audit is to write the packing and product-Dirichlet-kernel row-sum proof with every logarithmic and content factor explicit.

Part V. Smith Form, Prime-Ideal Extraction, and Recursion

19. Explicit strip coefficients

PROVED: The coefficient matrix has determinant $N(u_2)$ and Smith form $\text{diag}(1, N(u_2))$ for primitive u_2 .

$$P=Ag-Bf; Q=A(f-g)+Bg; \text{index}(d|P,Q)=d^2/\gcd(d, N(u_2)).$$

20. Large regular content

PROVED: If $\gcd(q, N(u_2))=1$, then qR divides u_1v_1 .

Invertibility modulo q forces both Eisenstein coordinates of u_1v_1 to be divisible by q . Unique factorization allocates (q) between u_1 and v_1 and reduces the product scale from X to X/q^2 .

21. Singular prime powers

PROVED: Singular Smith content has an asymmetric prime-ideal interpretation.

Let p^a exactly divide $N(u_2)$, p not equal to 3, and choose $p=\pi \cdot \bar{\pi}$ so that π^a divides u_2 and $\bar{\pi}$ does not. If p^k divides q , then $\pi^{\max(k-a,0)} \bar{\pi}^k$ divides u_1v_1 . Its norm is $p^{(2k-\min(k,a))}$, exactly the local Smith index.

22. Unified extracted ideal

PROVED: There is an extracted ideal $G(q, u_2)$ dividing u_1v_1 with norm $q^2/\gcd(q, N(u_2))$, up to bounded ramified-prime bookkeeping.

$$N(G)=D(q, n)=q^2/\gcd(q, n), \quad n=N(u_2).$$

Hence every large-content block $q > V^{1/4}$ descends the product scale by at least $V^{1/4}$.

23. Global allocation recurrence

AUDIT: The large-content recurrence closes once the small-content additive sieve is independently verified.

Allocate $G=g \cdot h$ with $g|u1$ and $h|v1$. Divisor-bounded allocation multiplicity and restricted L2 norms preserve a factor $1/N(G)$. Summing extracted ideals with norm above Q gives a $Q^{-1} X^\epsilon$ contraction.

$$C_{\text{off}}(X) \leq C_0 + O(X^\epsilon/Q) C_*(X/Q), \quad Q=V^{1/4}.$$

24. Exact diagonal

PROVED: Weighted multiplicative energy is X^ϵ -bounded uniformly for unequal U,V .

$$E_x(a,b) \ll X^\epsilon ||a||_2^2 ||b||_2^2.$$

This controls the M-term. The proof uses the exact parameterization $u1=gr, u2=gs, v1=sh, v2=rh$ with $(r,s)=1$ and two applications of Cauchy-Schwarz followed by Eisenstein divisor bounds.

Part VI. Current Synthesis

Component	Status
Modulo-6 tracks and sign algebra	PROVED
360 wheel	MODEL plus exact residue bookkeeping
Eisenstein norm/determinant/UFD	PROVED
$Q(q,r)$ involution	PROVED
Finite prime/lattice scans	COMPUTED
Smith index	PROVED
Singular prime-ideal extraction	PROVED
Weighted exact diagonal energy	PROVED
Full 2-D additive-sieve bookkeeping	AUDIT
Full Type-II angular estimate	CONDITIONAL on audit
RH/operator conclusion	CONDITIONAL; not established

The hierarchy is now clear: fixed residue arithmetic is foundational; the Eisenstein lattice supplies norms, ideals, arguments, sixfold symmetry, and determinant geometry; wheel/cone/sphere pictures are intuition layered above that arithmetic; and the analytic layer studies angular characters and bilinear sums while preserving factorization.

The principal remaining verification is the complete two-dimensional additive large-sieve proof with primitive-frequency reduction, D^{-2} packing, product Dirichlet-kernel row sums, logarithmic losses, denominator collapse under rational content, smooth norm coupling, and the ramified prime 3. Until that is checked line by line, the final Type-II theorem remains conditional.

Appendices: Proofs and Technical Derivations

Appendix A. Modulo-6 track algebra

If $p > 3$ is prime, p is not divisible by 2 or 3. The reduced residue classes modulo 6 are 1 and 5, so $p = 6n+1$ or $p = 6n-1$. Expanding $(6a+s)(6b+t)$ gives $36ab+6at+6bs+st=6(6ab+at+bs)+st$. Since s and t are plus or minus one, the output sign is again plus or minus one. The multiplication table of the two nonzero residue classes proves multiplicativity of χ_6 . Multiplying by $x^{-1/2}$ then proves multiplicativity of W on integers coprime to 6.

Appendix B. Eisenstein norm and determinant-angle identity

Let $\omega = (-1+i\sqrt{3})/2$. Conjugation sends ω to ω^2 . Direct multiplication gives $N(a+b\omega) = a^2 - ab + b^2$. For $z = a+b\omega$ and $w = c+d\omega$, direct expansion gives $\text{Im}(z \text{ conjugate}(w)) = -(\sqrt{3}/2)(ad-bc)$. Since $\text{Im}(z \text{ conjugate}(w)) = |z||w| \sin(\arg z - \arg w)$, taking absolute values proves the determinant-angle identity.

Appendix C. Invariance of the quadratic cone form

For $T(q,r) = (q, -3q-r)$, applying T twice gives (q,r) . Substitution into Q gives $3q^2 + 3q(-3q-r) + (-3q-r)^2$. Expanding and collecting terms yields $3q^2 + 3qr + r^2 = Q(q,r)$. Thus T is an involutive isometry of Q .

Appendix D. Primitive/content decomposition

For a nonzero Eisenstein ideal a , let d be the largest positive rational integer for which (d) divides a . Then $a = (d)a_0$ and a_0 has no further rational integer divisor greater than one. Maximality makes d unique, and cancellation in the integral domain makes a_0 unique. For elements, the same statement holds after choosing normalized generators; units create only a fixed sixfold ambiguity. Divisor sums over d are harmless in dyadic ranges because the Eisenstein divisor function is N^ϵ -bounded.

Appendix E. Strip coefficients and Smith normal form

Write $\alpha = A+B\omega$, $u_2 = f+g\omega$, and $v = x+y\omega$. Multiplication in $Z[\omega]$ and the determinant definition give $[\alpha, u_2 v] = Px + Qy$ with $P = Ag - Bf$ and $Q = A(f-g) + Bg$. Therefore $(P, Q)^T = T(A, B)^T$ with $T = \begin{bmatrix} g & -f \\ f-g & g \end{bmatrix}$. Its determinant is $f^2 - fg + g^2 = N(u_2)$. If $\gcd(f, g) = 1$, the gcd of the one-by-one minors is one, so the Smith invariants are 1 and $N(u_2)$. Consequently the congruence $d|P, Q$ has index $d^2/\gcd(d, N(u_2))$.

Appendix F. Regular-content extraction

Assume q divides P and Q and $\gcd(q, N(u_2)) = 1$. The strip matrix is invertible modulo q , so A and B are both zero modulo q . Hence qR divides $\alpha = u_1 v_1$. Factor the ideal (q) into prime ideals and distribute each prime-power exponent between u_1 and v_1 according to their valuations. Since $Z[\omega]$ is a PID, this yields $u_1 = \gamma a$ and $v_1 = \delta b$ with $(\gamma)(\delta) = (q)$, up to a unit. Therefore the residual product ab has norm X/q^2 . The number of allocations is bounded by the Eisenstein divisor function of (q) .

Appendix G. Singular prime-power extraction

Let u_2 be rationally primitive and let p not equal to 3 divide $N(u_2)$. An inert p would divide both rational coordinates of u_2 , contradicting primitivity, so p splits as $\pi \pi_{\bar{}}$. Primitivity also prevents both π and $\pi_{\bar{}}$ from dividing u_2 . Choose the labeling so $v_{\pi}(u_2) = a$ and $v_{\pi_{\bar{}}}(u_2) = 0$. If p^k divides both strip coefficients, the $\pi_{\bar{}}$ component is nonsingular and forces $\pi_{\bar{}}^k$ to divide α . At π , multiplication by u_2 already supplies a powers, so α must supply $\max(k-a, 0)$ additional powers. Hence $\pi^{(\max(k-a, 0))} \pi_{\bar{}}^k$ divides α . Its norm is $p^{(2k - \min(k, a))}$, equal to the Smith index. Multiplying these local extracted ideals gives $N(G) = q^2/\gcd(q, N(u_2))$, with the ramified prime 3 handled separately as a bounded local exception.

Appendix H. Weighted exact multiplicative energy

Consider $u_1 v_1 = u_2 v_2$ with $N(u_i) \sim U$ and $N(v_i) \sim V$. Let $g = \gcd(u_1, u_2)$, write $u_1 = gr$ and $u_2 = gs$ with $\gcd(r, s) = 1$. Then $rv_1 = sv_2$. Unique factorization implies s divides v_1 and r divides v_2 , so $v_1 = sh$ and $v_2 = rh$. Thus every solution has the form (gr, sh, gs, rh) .

Define $A(r)=\sum_{\{r|u\}}|a_u|^2$ and $B(r)=\sum_{\{r|v\}}|b_v|^2$. Pointwise Cauchy-Schwarz bounds the r,s correlation by $\sqrt{A(r)A(s)B(r)B(s)}$. Summing and applying Cauchy-Schwarz again gives energy at most $(\sum_r A(r))(\sum_r B(r))$. Interchanging divisor and coefficient sums gives $\sum_r A(r)=\sum_u |a_u|^2 \tau_R(u)$, and similarly for B . The Eisenstein divisor bound $\tau_R(z) \ll N(z)^\epsilon$ yields $E_X \ll U^\epsilon V^\epsilon \|a\|_2^2 \|b\|_2^2 = X^\epsilon \|a\|_2^2 \|b\|_2^2$. No assumption $U \sim V$ is used.

Appendix I. Centered local mean

Let ψ_s be a smooth bounded-overlap angular partition on a fundamental sector I of length L , and let ρ_r be a radial partition. Define $A_{\{r,s\}}=\sum_u c(u)\rho_r(Nu)\psi_s(\arg u)$, $M_r=\sum_s A_{\{r,s\}}$, and $\kappa_s=L^{-1} \int_I \psi_s$. The local uniform mean is $\kappa_s M_r$ and the centered coefficient is $A^o_{\{r,s\}}=A_{\{r,s\}}-\kappa_s M_r$. Because $\sum_s \kappa_s=1$, $\sum_s A^o_{\{r,s\}}=0$. Translation-invariant angular kernels then split the incidence exactly into a bulk uniform-density term and a centered fluctuation. A short-sector coefficient example shows that centering alone does not imply the desired arbitrary-coefficient fourth-moment bound.

Appendix J. Two-dimensional additive large sieve: audited core and remaining detail

For reduced frequencies $\xi_i=(r/d,s/d)$ with $d \leq D$, distinct torus points are separated in the sup norm by at least D^{-2} . An Eisenstein annulus $N(z) \sim V$ lies in a box of side $O(\sqrt{V})$. By duality, the additive large sieve reduces to a row-sum estimate for products of one-dimensional Dirichlet kernels. Packing the frequency points into D^{-2} squares leads to the expected scale $(\sqrt{V}+D^2)^2$, which is bounded by a constant times $V+D^4$. Logarithmic losses from dyadic annuli in frequency space are harmless as X^ϵ .

What remains to be written independently and checked line by line is the full row-sum argument with primitive frequency reduction, duplicate rational representations, rational-content denominator collapse, Smith multiplicity, smooth product-norm coupling, and the ramified prime 3. This appendix therefore records the analytic core but deliberately labels the complete bookkeeping as an audit rather than a finished theorem.

Appendix K. Large-content recurrence

Let $D(q,n)=q^2/\gcd(q,n)$ and let G be the extracted ideal of that norm. Allocate $G=g*h$ with $g|u1$ and $h|v1$. For each allocation, the residual Type-II scale is X/D . The number of allocations is $\tau_R(G) \ll D^\epsilon$. Expanding the restricted $L2$ norms and summing over allocations shows that the large-content contribution contains a factor $1/D$. Summing extracted ideals with $D>Q$ gives at most $Q^{-1} X^\epsilon$ after divisor counting. With $Q=V^{1/4}$ and $V \geq X^\epsilon$, this is a power-saving contraction. Therefore the large-content part closes by induction once the small-content additive estimate is available.

Appendix L. Proof-status ledger

Statement	Status
Prime tracks $6n+/-1$	PROVED
Signed-track multiplication	PROVED
χ_6 and W multiplicativity	PROVED
Inverse-limit residue topology	PROVED
Eisenstein norm/determinant identity	PROVED
Q involution	PROVED
Global cone-to-track bridge	OPEN
Centered mean decomposition	PROVED
Same-sequence centered fourth-moment target	DISPROVED for arbitrary coefficients
Smith index	PROVED

Statement	Status
Regular q extraction	PROVED
Singular prime-ideal extraction	PROVED
Weighted exact diagonal energy	PROVED
Complete small-content additive sieve bookkeeping	AUDIT
Full Type-II angular theorem	CONDITIONAL on audit
Unconditional RH conclusion	NOT ESTABLISHED

Appendix M. Computational reproducibility checklist

Future numerical sections should record the exact coordinate map, scan range, primality test, treatment of units and associates, random seeds for probabilistic tests, definition of every statistic, and source code or pseudocode sufficient for independent replication. Reported finite observations should remain labeled computational even when the match rate is 100 percent over the tested range.

Conclusion

The Trinary Symmetry project has evolved from a visually driven modular model into a layered arithmetic and analytic program. Several early geometric intuitions have survived in corrected algebraic form: antipodal tracks became a modulo-6 character, sixfold wheel symmetry became Eisenstein-unit symmetry, boundary geometry became determinant-strip content, and singular strip degeneracy became a prime-ideal extraction theorem. Other ideas remain models or open bridges, especially the global cone-coordinate labeling and the original operator/RH conclusion.

The strongest current analytic program is the bilinear Eisenstein Type-II approach. Its exact diagonal, content algebra, Smith form, regular and singular extraction, and large-content recursion now have explicit proofs. The remaining decisive task is an independent, fully detailed verification of the small-content two-dimensional additive large-sieve estimate with all bookkeeping. That separation of proved structure from remaining audit is the basis on which the project can now be presented for serious mathematical review.